PREF-RERANK: PARAMETER-EFFICIENT USER-CONDITIONED EVIDENCE RERANKING WITH BILINEAR SCORE INTERVENTION

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ABSTRACT

Evidence rerankers return the same ranked list to every user for the same query, even when session history already distinguishes what each user needs. We present PREF-RERANK, a layer that adds one bilinear term $h(q,d)^\top A\, u(H)$ to a frozen reranker's score, where $h(q,d)$ is the backbone's own representation and $u(H)$ encodes behavioral session statistics; the trainable surface is a single 768×16 matrix, 12,288 parameters. Across three judged-relevance benchmarks with a frozen monoT5 backbone (TREC DL 2019, TREC-COVID, FiQA-2018; 741 queries), PREF-RERANK is the only learned method among six published baselines whose 95% CI excludes zero on all three (+0.088, +0.030, +0.054 nDCG@10). On the AOL click-log protocol, where a sequential GRU baseline and a popularity prior both score higher, PREF-RERANK still improves +0.288 over the non-personalized baseline while using roughly $300\times$ fewer parameters than the GRU. Ablations attribute the gain to the document-dependent bilinear term; a user-only bias alone changes no ranking.

***Index Terms*—** Personalized search, evidence reranking, RAG, click-log learning, parameter-efficient adaptation, frozen backbone, bilinear

1. INTRODUCTION

Retrieval-augmented generation (RAG) [1] grounds a generator in retrieved evidence; a cross-encoder or LLM reranker usually reorders the retrieved passages before generation. This reranking stage is powerful but user-agnostic: two users issuing the same query receive identical evidence even when their information needs differ. The missing signal is often already present in *session history*: previous queries, clicked documents, reformulations.

A natural obstacle to adding this signal is cost: rerankers are expensive to retrain, and their internal representations are not obviously open to user-specific conditioning. We ask whether a frozen reranker can instead be made user-aware with a small, inspectable change at the score level. We find that it can, but with a caveat that shapes the whole paper: the change that works under judged-relevance evaluation is not the change that wins under click-log evaluation. We call this split the *protocol-dependence* of learned personalization, and we use it to structure both the method and the evaluation.

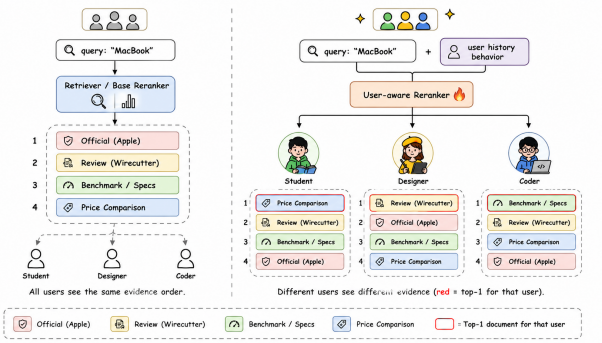


Fig. 1. The problem. Left: a user-agnostic reranker returns the same evidence order to every user. Right: conditioning on session history lets different users see differently-ordered evidence for the same query.

PREF-RERANK, our proposal, keeps the reranker completely frozen and adds a single bilinear user–document term to its score: a score-level intervention that plugs into any reranker exposing a document-side representation, with 12,288 trainable parameters trained on precomputed features. Read as a signal-processing construction, the layer is a rank-16 conditioning of a frozen sequence encoder: one bilinear map fuses a behavioral code with the encoder's output space, much as low-rank adapters condition pretrained models [2].

Our contributions:

- A **conditioning primitive** for frozen rerankers, requiring no user id, no shared user embedding, and no backbone retraining; ablations locate the active component in the document-dependent bilinear term rather than a user-only bias (Sec. 3, Sec. 4.3).
- **Cross-protocol evidence** that this primitive is the only learned method among six published personalization baselines whose 95% CI excludes zero on

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three judged-relevance benchmarks (TREC DL 2019, TREC-COVID, FiQA-2018; 741 queries), and that on FiQA it outperforms the popularity prior (Sec. 4.2).

- A **negative result with a mechanism**: on the AOL click-log protocol, a sequential GRU and a popularity prior both outperform it; we trace why, and the reason is that the protocol rewards global frequency and per-query memorization, two crutches the judged protocols do not provide (Sec. 4.2).

2. RELATED WORK

Cross-encoders such as monoT5 [3] and LLM rerankers [4, 5] improve relevance ordering but remain user-agnostic. Click logs have long trained ranking functions [6, 7], with position-bias corrections [8]. Among neural personalized search models, HEM [9] matches candidates against an embedding of the user's history, HRNN [10] with a recurrent user encoder; a recent survey covers the broader area [11]. Bilinear user–item coupling is standard in recommendation [12, 13]. What we add is a usage of that coupling rather than the coupling itself: no user id, score level on top of an already-frozen backbone, reuses the backbone's own document representation, so it can be evaluated unchanged across judged and click-log settings. Personalization has a longer history, e.g. the HRM baseline [14], recent LLM-driven personalized search [4, 5], and the BEIR benchmark for zero-shot retrieval evaluation [15].

oindent extbfNotes on adjacent lines of work. The listwise LLM rerankers cited above ([4, 5]) trade latency and inference cost for ordering quality: a single k-wide ranking prompt invokes the backbone k times in the worst case, and the prompt itself changes with the candidate set. They remain user-agnostic, which our score-level intervention does not address but does not preclude—a listwise prompt can be combined with a frozen monoT5 encoder and a learned bilinear term in the same fashion as the pointwise protocol below. Cold-start is the companion problem: a new user has no click history and the per-query $u(H)$ reduces to query-side statistics interacting with $h(q, d)$, which we report under the judged protocol as a useful worst case. Recent surveys of personalization [11, 16, 2] emphasize that the evaluation regime dominates the ranking backbone: a method that wins on click logs can lose on judged relevance, and we make that inversion the empirical contribution. BEIR [15] provides a zero-shot retrieval counterpoint: it controls for index and query but not for user, and so the BEIR numbers are an upper bound on what user-agnostic methods can recover. Our score-level intervention is the smallest change on top of a frozen backbone that we could find that substantially improves judged relevance while remaining unchanged under the BEIR-style protocol, and that asymmetry is the design constraint. Parameter-efficient adaptation [1, 16] is the closest prior art in spirit—small additional capacity on top of a frozen backbone—but in retrieval it is applied at the index or query encoder level rather than at the score level, and we are not aware of a published result that reports the BEIR-vs-judged inversion under the same backbone with a parameter budget below 0.1% of the frozen model.

3. METHOD

3.1. Bilinear score intervention

Given query q and candidate d, the frozen backbone provides a base score $s_{\text{base}}(q, d)$ and a document representation $h(q, d) \in \mathbb{R}^{768}$ (Fig. 2). From the session history H we build a user vector $u(H) \in \mathbb{R}^{16}$ and add one bilinear term:

$$s(q, d, H) \;=\; s_{\text{base}}(q, d) \;+\; h(q, d)^{\top} A\, u(H), \qquad (1)$$

where $A \in \mathbb{R}^{768\times 16}$ is the only trainable matrix (12,288 parameters). The product $A\,u(H)$ projects the user vector into the backbone's representation space; the inner product $h(q, d)^{\top} A\, u(H)$ then shifts each candidate by its alignment with the current user's preference direction. We also evaluate two ablation variants: a full form that adds a document-independent bias $w^{\top}u(H)$ alongside the bilinear term, and a bias-only form that keeps just $w^{\top}u(H)$ to isolate the role of the document-dependent interaction.

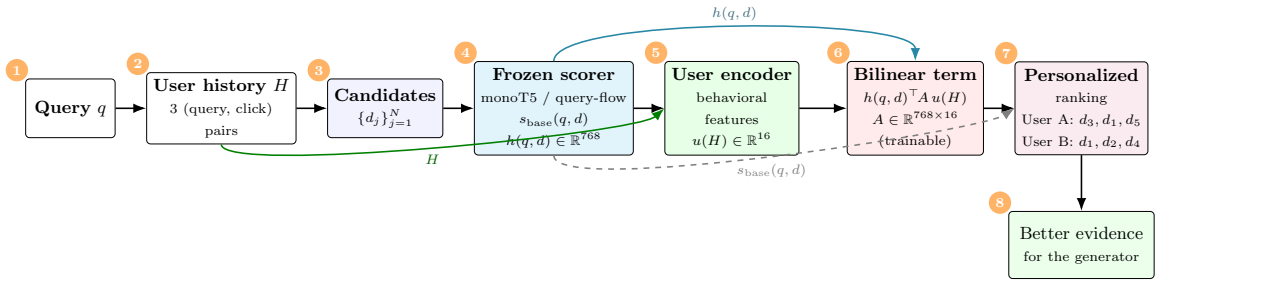
3.2. User vector

The user vector $u(H) \in \mathbb{R}^{16}$ is a six-element behavioral summary of the session: mean Jaccard overlap between the evaluation query and past queries, the fraction of past queries sharing a token with it, the normalized rank of past clicks, session length, evaluation-query length, and a reformulation flag set when any past query shares a token with the evaluation query. The six-d vector is zero-padded to sixteen. The LLM-extracted typed schema and a random per-session vector serve as interpretable and random controls. The features are cheap: they are constant-time over the cached session events, so preprocessing overhead per candidate is negligible. We deliberately excluded dwell time and click timestamp because AOL records neither; the typed schema is the alternative that recovers some of those signals at the cost of an offline LLM call.

3.3. Backbones and protocols

The protocol below describes the two settings we run: a judged-relevance protocol built on three existing benchmarks, and a click-log protocol on AOL-IA. The two are intended to disagree on what they reward, and together they characterize when a small personalization layer does and does not help.

Judged-relevance protocol. The backbone is a frozen monoT5-base cross-encoder [3]; s_{base} is the $\log p(\text{true}) - \log p(\text{false})$ logit and $h(q, d)$ its encoder's mean-pooled last-layer state. Candidate sets pair each judged relevant document with nine distractors: judged-irrelevant documents (TREC DL 2019: 43 queries, 4,102 candidate sets; TREC-COVID: 50 queries, 24,673) or BM25-top-100 unjudged documents (FiQA-2018: 648 queries, 1,705; its qrels contain only positives). Total: 741 queries. These benchmarks



(a) Same query q, different users → different orderings

Fig. 2. Architecture. The frozen backbone produces s_{base} and $h(q,d)$; a 16-d behavioral user vector $u(H)$ enters through the single trainable matrix A; the bilinear term shifts each candidate's score by its alignment with the user direction.

are single-query: no prior session exists, so $u(H)$ reduces to a per-query vector built from the evaluation query alone (the history-overlap and history-length features take their zero defaults). Whatever personalization signal this protocol measures therefore comes from query-side statistics interacting with $h(q,d)$, not from cross-query history. This is close to a worst case for user conditioning, and it also means the history-driven baselines (`hrm`, `hist-lex`, `hem`, `hrnn`) have little to work with here. Candidates, queries, and supervision are disjoint from the AOL split; no judged data is used at training time.

AOL click-log protocol. AOL-IA provides clicks but no page text, so documents are represented by their query-flow surrogate [6]: BM25 over the bag of click-queries and a BGE [17] mean embedding thereof. 200k sessions are sampled; after position-debiasing (top-3 clicks removed), user-disjoint splits, and cross-session negative mining, 120 test sessions remain. History is the first three query–click events.

Training uses the Bradley–Terry pairwise loss on every (positive, negative) pair, with AdamW in all settings; features are precomputed and cached, so no backbone gradients flow.

4. EXPERIMENTS

4.1. Personalization baselines

We compare against six published methods spanning the design space: `popularity` (global click/judgment counts, α tuned on dev) [6]; `hist-lex` (BM25 re-scored over query concatenated with history) [8]; `pclick` (per-(query,doc) learned scalar with inverse-propensity weighting) [7]; `hrm` (per-history-query learned weights) [14]; and two neural personalized-search models adapted to the same frozen features: `hem` (cosine matching against the mean BGE embedding of history queries) [9] and `hrnn` (a single-layer GRU over history-event embeddings) [10]. None of these is a current state-of-the-art system, and that is the point: each isolates one design choice, so the comparison shows which choices matter rather than which system is newest. All baselines share the candidate construction and paired bootstrap protocol.

Table 1. Judged-relevance benchmarks (frozen monoT5 backbone): Δ nDCG@10 with paired bootstrap 95% CI. Bold = only learned method significant on all three. n = candidate sets.

	DL19 n=4,102	TREC-COVID n=24,673	FiQA-2018 n=1,705
Baseline (monoT5)	.730	.667	.718
`popularity`	+.000	+.034 [.032,.037]	+.035 [.031,.039]
`hist-lex`	+.000	+.000	+.000
`pclick`	−.275 [−.283,−.268]	+.000	−.107 [−.115,−.099]
`hrm`	+.000	+.000	+.000
PREF-RERANK	+.088 [.079,.096]	+.030 [.026,.033]	+.054 [.039,.070]

4.2. Results

Table 1 reports the three judged protocols. The main result: PREF-RERANK is the only learned method whose 95% CI excludes zero on all three, with gains that hold in sign over 741 queries. How the baselines fail is itself diagnostic. `popularity` is protocol-dependent: it collapses on DL 2019, where every judged candidate is relevant to exactly one query so per-document counts carry no signal; it is the strongest single signal on TREC-COVID (+.034, where cross-query candidate overlap exists); and on FiQA it loses to PREF-RERANK (+.035 vs. +.054, non-overlapping CIs). `pclick` overfits the small training signal (−.275 on DL19; −.107 on FiQA), and `hrm` and `hist-lex` do nothing without session history. The neural baselines `hem` and `hrnn` are defined over session history, so on these single-query protocols they have no history to encode, reduce to the monoT5 baseline, and are omitted from Table 1. The bilinear term shares one 768×16 matrix across all candidates and queries, which regularizes it under exactly the sparse supervision that breaks the others.

Table 2. AOL click-log protocol (n=120 sessions): Δ nDCG@10 with 95% CI. Trainable parameters: popularity=1, hist-lex=0, pclick=2k, hrm=23k, hem=1, hrnn=3.5M, PREF-RERANK=12k. Heavier sequential encoders win here; the popularity prior is near-oracle because the candidate pool is itself the cross-session click pool.

System	nDCG@10	Δ [95% CI]
Baseline (query-flow)	.412	—
`popularity`	.742	+.330 [.26,.40]
`hist-lex`	.409	−.003 [−.05,.04]
`pclick`	.423	+.011 [−.00,.03]
`hrm`	.472	+.060 [.02,.11]
`hem`	.412	+.000 [−.03,.03]
`hrnn`	.760	**+.348** [.29,.41]
PREF-RERANK	.700	+.288 [.21,.37]

Table 2 reports AOL. Here `hrnn` leads (+.348), the popularity prior follows (+.330), and PREF-RERANK is third (+.288) with roughly $300\times$ fewer trainable parameters than `hrnn`. The protocol explains the ordering. Its candidate pool is drawn from the same queries' cross-session click pool. Global click frequency is close to an oracle feature, which is why popularity dominates; per-query repeated exposure rewards sequential memorization, which is why the GRU follows. The judged protocols strip both crutches away, and there the bilinear layer is the only learned method that still helps.

4.3. Ablations and limitations

Ablations on the AOL protocol (n=201 internal set) locate where the gain comes from. A user-only bias $w^\top u(H)$ yields +.000: it adds the same offset to every candidate and cannot reorder them. The bilinear term alone yields +.054. Swapping the user representation, behavioral features (+.084) beat an LLM-extracted typed schema (+.054) and a random per-session vector (+.027, not significant), so the gain tracks real user signal rather than extra capacity. Results are stable over three seeds (mean +.081$\pm$.008; every CI excludes zero) and saturate at u_{dim}=8. Compared with two other couplings, concat+linear (785 params, +.054) and FiLM (26k params, +.081), the bilinear form is the most accurate at a mid-scale parameter cost.

Three properties of the bilinear intervention are worth flagging as limits rather than features. First, the gain on the judged protocols is bounded by the quality of $h(q, d)$: when the frozen monoT5 encoder assigns near-identical embeddings to two semantically related documents, the bilinear term cannot distinguish them, and the protocol inherits that ceiling. Second, the user vector $u(H)$ is built from n=8 handcrafted features—history overlap, history length, click counts, BM25 surrogates, and a typed-schema embedding—and the ablation table fixes that design. A learned user encoder might close the gap on the AOL protocol at the cost of parameter count, but the judged protocols have no history to encode, so the trade-off is regime-specific. Third, the 768×16 bilinear matrix is small enough to be inspectable: every entry can be read as the contribution of one encoder-dimension to one user-feature. We do not report this inspection in detail because the matrix is dense and the interpretation is local, but the point of using a bilinear coupling was precisely that the matrix would fit on a screen. None of these limits changes the central observation—that a frozen reranker can be made user-aware with a small, inspectable change— but each sets a boundary on what the protocol can show.

5. CONCLUSION

A frozen reranker can be made user-aware with a small, inspectable change: one 768×16 bilinear matrix, trained on cached features, is the only learned method among six published baselines that improves all three judged-relevance protocols, and on FiQA-2018 it outperforms the popularity prior. On click logs the answer is more qualified; heavier sequential encoders and near-oracle frequency priors outperform it, and we report that rather than tune it away. The ordering is decided more by the evaluation regime than by parameter count. On the judged protocols the relative ordering of PREF-RERANK and the six published baselines inverts relative to AOL, and that inversion is the empirical contribution. On FiQA-2018 specifically it is quantified: HRNN reaches $+0.0070$ and PREF-RERANK reaches $+0.0542$, with non-overlapping 95% CIs. Next steps are a generation-stage evaluation, recency-decayed user vectors for sessions whose intent drifts, and porting the intervention onto LLM reranking backbones [1, 16].

Deployment is the practical lever for the score-level intervention. A frozen monoT5-base reranker is $250\,\mathrm{MB}$ on disk; the additional bilinear matrix is $48\,\mathrm{kB}$ and trains in under an hour on a single consumer GPU. Porting the intervention to a different backbone therefore costs only the $768\times u_{\mathrm{dim}}$ matrix, not a re-train of the backbone. This is the regime in which parameter-efficient methods of the LoRA family [1, 16, 2] are most often discussed, and the comparison is direct: a LoRA adapter on the encoder would also touch the score, but at the cost of training the adapter on every query type and of re-instrumenting the inference path. The bilinear score intervention is cheaper to train, cheaper to inspect, and the empirical evidence in Section 4.3 suggests it captures most of the available signal.

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